

Jawaharlal Nehru University, New Delhi-10067
School of Computer & Systems Sciences
Mid Semester Examination-MCA Semester (II)
(CS-407) Object Oriented Programming

Duration: 1 hour 30 minutes

Max Marks: 25
Date: 20/05/2023

All questions are compulsory

Q.1 Attempt the followings[5 x 2]

[i] Consider the following program

```
public class Test {  
    public static void main(String[] args) {  
        /***** boolean b = false; *****/ //n1  
        String b = "false";  
        switch(b){ // n2  
            case "False":  
                System.out.println("a");  
        }  
    }  
}
```

What is the output of the above code? Also Justify your Answer.

- a. a
- b. Compiler error due to line n1
- c. Compiler error due to line n2
- d. Print nothing

[ii] Following is a piece of code where some parts of a statement is missing:

```
public class Question{  
    public static void main(String args[]){  
        char npitel[]={ '1', '2', '3', '4', '5', '6' };  
        System.out.print( _____ );  
    }  
}
```

In the following, some options are given. You have to choose the correct option(s) for the argument in *System.out.print()* function to print the ASCII value **102**.

What is the output of the above code? Also Justify your Answer.

- a. `npitel[npitel.length-2] + npitel[0]` ~~x~~
- b. `npitel[0] + npitel[npitel.length-2]` ~~x~~
- c. `"" + npitel[npitel.length-2] + npitel[0]`
- d. `"" + npitel[0] + npitel[npitel.length-2]`

[iii] Following is a program given for this question.

```
public class Question{  
    public static void main(String[] args){  
        int[] x = {222, 210, 012};  
        for(int i = 0; i < x.length; i++){  
            System.out.print(x[i] + "");  
        }  
    }  
}
```

What will be the output of the above program? Also Justify your Answer.

- a. 22221010
- b. 12222101
- c. 22101010
- d. 22221012

[iv] Consider the following program and identify the output.

```
public class Question{
public static void main(String[] args){
short x = 10;
x = x * 5;
System.out.print(x);
}
}
```

What will be the output of the above program? Also Justify your Answer.

- a. 5
- b. 10
- c. 50
- d. Compilation error

[v] Consider the following piece of code.

```
public class Question{
Question() {
this();
}
public static void main(String []args) {
Question obj = new Question();
System.out.println("Java");
}
}
```

Which of the following is the output of the above program? Also Justify your Answer.

- a. Java
- b. There will be a compile-time error.
- c. JavaJava.
- d. The program will give a runtime error.

Q.2 A large Indian Petroleum company has created a number of cylindrical tanks on a rectangular field. External surfaces of these are to be painted, including the flat circular cover on the top. Dimensions and number of tanks are given as input (use scanner method), for a given price of painting work per Sq. meter, and Find the cost of painting the tanks. Write a Java Program for the same. [4]



Q.3 Write a Java Program for creating two classes namely class1 for Circle & class 2 for Box. Using Methods declare and initialize the objects of class type Circle and Box (Encapsulation). For class 1 display the circumference and area of the circle & for class 2 display area and volume of the box as an output. [5]

Q.4 Write a Java Program using constructor overloading method (Polymorphism) to define class Room with two constructors one for rectangle room & other for square room also display the 'area' of the same.[6]